

Ravada Koteswar Rao

Mechanical Design Engineer

✉ koteswarraoravada@gmail.com ☎ 7997004883 📍 Visakhapatnam, Andhra Pradesh, India

PROFILE

Mechanical Design Engineer with 2+ years of experience in industrial equipment design, reverse engineering, and engineering support for Tata Steel projects. Proficient in Siemens NX, PTC Creo, AutoCAD, GD&T, Design for Manufacturability (DFM), and Design for Assembly (DFA). Experienced in developing 3D component and assembly models, manufacturing drawings, fabrication drawings, BOMs, and engineering documentation. Strong understanding of machine design fundamentals, manufacturing processes, and design validation through site engineering and client collaboration

PROFESSIONAL EXPERIENCE

Praxsol Engineering Private Limited

Junior Design Engineer

Clients : TATA Steel, MVS ACMEI

03/2024 – Present

Visakhapatnam,

Andhra Pradesh

Design Engineering Activities

- Developed 200+ manufacturing and fabrication drawings for mechanical components and assemblies used in industrial plant equipment.
- Created accurate 3D component and assembly models using Siemens NX and PTC Creo and developed 2D drawings using AutoCAD.
- Prepared manufacturing, fabrication, and General Arrangement (GA) drawings for production and installation activities.
- Generated and maintained Bills of Materials (BOM), ensuring proper documentation and traceability.
- Applied GD&T, Design for Manufacturability (DFM), and Design for Assembly (DFA) principles to improve manufacturability and assembly feasibility.
- Performed reverse engineering by converting site measurements, sketches, and existing components into production-ready CAD models and drawings.
- Collaborated with clients and plant personnel in technical discussions and engineering reviews.

Site Engineering Activities

- Worked on-site at Tata Steel Jamshedpur and Tata Steel Kalinganagar supporting engineering and design activities.
- Performed reverse engineering through accurate on-site measurements and detailed hand sketches for design development.
- Coordinated with plant engineers and stakeholders to resolve technical issues and implement design solutions.
- Led and guided a team of four Graduate Engineer Trainees, ensuring quality and timely delivery of design outputs.
- Reviewed GA drawings and critical engineering data through discussions with clients and on-site personnel to support accurate design development.
- Ensured accurate alignment of drawings with actual plant conditions through on-site verification, measurements, and installation feasibility checks.

SKILLS

CAD & PLM Tools

- Siemens NX CAD
- PTC Creo Parametric
- AutoCAD
- CATIA V5
- Teamcenter (Self-Learning)

Standards & Documentation

- GD&T Fundamentals
- Indian Standards (IS)
- Bill of Materials (BOM) Creation & Documentation
- Design for Manufacturability (DFM)
- Design for Assembly (DFA)

Design & Engineering

- Reverse Engineering
- 3D CAD Modeling
- Manufacturing & Fabrication Drawings
- General Arrangement (GA) Drawings

EDUCATION

Bachelor of Technology Anil Neerukonda Institute of Technology and Sciences (Mechanical Engineering) Cumulative GPA : 8.48	2021 – 2024 Visakhapatnam, Andhra Pradesh
Diploma in Mechanical Government Polytechnic Visakhapatnam Percentage : 79.23%	2018 – 2021 Visakhapatnam, Andhra Pradesh
Board of Secondary Education, AP M.P. & E.V. E.M. High School Cumulative GPA : 9.2	2017 – 2018 Visakhapatnam, Andhra Pradesh

PROJECTS

Reverse Engineering of Mechanical Components (End-to-End Workflow) - Executed complete reverse engineering workflow from site measurements and hand sketches to manufacturing drawings. - Developed detailed drawings for shafts, gears, gearboxes, pulleys, sprockets, chains, plummer blocks, springs, trolleys, base frames, and sheet metal components. - Ensured dimensional accuracy, fitment, and manufacturability.	04/2024 – Present
Web Centric Design for Smart Gate Systems Developed a web-based system for smart gate design, enabling efficient design iterations and usability with integrated mobile control and monitoring.	12/2023 – 03/2024

CERTIFICATES

Siemens NX Linkedin Learning
CATIA V5 (CAD) Infosys Springboard.
Introduction to Geometric Dimensioning and Tolerancing (GD&T) Linkedin Learning
Additive Manufacturing Manufacturability : Laser Powder Bed Fusion(L-PBF) American Society of Mechanical Engineers (ASME).

PERSONALITY TRAITS

- Problem-solving skills
- Team player
- Fluent communication skills

LANGUAGES

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| • English | • Hindi | • Telugu |
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